

1. [Dis] (5 pts) What types of “Learning Blocks” are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self-supervised learning, etc.)?

For each type, provide at least one example model that can be implemented using the free tier.

- Supervised - Classification - Mask detection
 - Example: Classify camera images as “mask” vs. “no mask.”
- Supervised - Regression - Temperature prediction
 - Example: Predict a continuous temperature value from sensor readings.
- Unsupervised - Anomaly Detection (GMM) - Fan anomaly detection
 - Example: Train on normal fan vibration/audio data and detect abnormal fan behavior.
- Unsupervised - Anomaly Detection (K-means) - Different normal operating modes of a machine
 - Example: Cluster normal machine behavior into different operating states, then flag unusual patterns.
- Visual Anomaly Detection (FOMO-AD) - Surface defect detection
 - Example: Detect scratches, cracks, missing parts, or other visual defects in images.
- Keyword Spotting (using Transfer Learning) - Voice command recognition
 - Example: Recognize simple commands such as “on,” “off,” “yes,” “no,” or a custom wake word.
- Object Detection (using MobileNetV2 SSD FPN) - Person/object detection
 - Example: Detect and locate people, tools, or parts in an image.
- Object Detection (using FOMO) - Counting small objects
 - Example: Detect and count objects such as screws, bottles, food items, or components on a table.
- Classical ML - Sensor-based classification or regression
 - Example: Use traditional machine learning methods to classify activities from accelerometer data or predict a sensor value.

2. [Dis] (5 pts) List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

Input layer - Receives the feature data from the processing block. It does not learn patterns by itself; it just passes the input features into the neural network.

- Dense layer - A fully connected layer where each neuron connects to the previous layer. It is used to learn patterns and relationships between features.
- Reshape layer - Changes the shape of the input data without changing the actual values. It is useful when the data needs to be rearranged before going into another layer.
- Flatten layer - Converts multi-dimensional data into a one-dimensional vector. It is usually used before a Dense layer.
- Dropout layer - Randomly turns off some neurons during training to reduce overfitting and make the model more generalizable.
- 1D Convolution layer - Extracts patterns from sequential data, such as audio, vibration, accelerometer, or other time-series signals.
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- 1D Pooling layer - Reduces the size of 1D feature maps after convolution. It helps lower computation and keeps the most important pattern information.
- 2D Convolution layer - Extracts spatial patterns from 2D data, such as images or spectrograms. It can learn features like edges, shapes, and textures.
- 2D Pooling layer - Reduces the size of 2D feature maps after convolution. It makes the model smaller and less sensitive to small location changes in the input.
- Output layer - Produces the final prediction, such as class probabilities for each label in a classification task.

3. [Dis] (5 pts) What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse? List all available methods.

- Quantized (int8)
- Unoptimized (float32)
- TensorFlow Lite
- EON Compiler
- EON Compiler (RAM optimized)

4. [Dis] (5 pts) What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications.

- Image modality - DALL-E Image Generation Block
 - Method: DALL-E model
 - Used to generate synthetic image datasets, such as objects, scenes, or visual classes.
- Audio/speech modality - Whisper Keyword Spotting Generation Block
 - Method: Whisper model
 - Used to generate keyword-spotting or speech-recognition datasets.
- Other possible synthetic data workflows - Physics simulation / NVIDIA Omniverse
 - Method: simulation-based generation
 - Used to create synthetic sensor, motion, or visual data from simulated environments.
- Synthetic data is important because it helps increase dataset size, add more variety, cover rare cases, and reduce the cost of collecting real-world data. It can also help improve model robustness when real data is limited.